
Calibrated Prompt Uncertainty in Zero-Shot Segmentation: Full-Validation Evidence from Lightweight Methods and SAM

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Abstract

1 Evaluating zero-shot prompted segmentation under uncertain input is difficult
2 because point and box perturbations have no directly comparable scale. We study
3 prompt robustness on all 1,449 PASCAL VOC 2012 validation images, using a
4 disjoint 100-image training subset to calibrate both channels to matched observable
5 quality. Center Color, GrabCut, and superpixel pipelines serve as interpretable CPU
6 baselines alongside SAM ViT-B. Three findings emerge. First, Robust Superpixel
7 improves over Center Color by 0.0640 IoU but remains below GrabCut, locating
8 its value in transparent component analysis rather than state-of-the-art accuracy.
9 Second, SAM’s score-based candidate selection provides a modest 0.0193 IoU
10 gain over a single noisy prompt. Third, matched-quality box perturbations become
11 increasingly more damaging than point perturbations as prompt quality decreases,
12 while the channels are indistinguishable near quality 0.9. Stratified results further
13 expose systematic failures on thin structures, small targets, and full-image boxes.
14 The study provides a reproducible robustness benchmark, component ablations,
15 and practical guidance for choosing prompt channels and segmentation methods.

16 1 Introduction

17 Interactive segmentation converts sparse user input into an object mask. Extreme points, iterative
18 clicks, boxes, and foundation-model prompts have progressively reduced the inference-time supervi-
19 sion requirement [4, 5, 6, 7]. Segment Anything (SAM) established a general point-and-box interface
20 with strong zero-shot transfer [7].

21 Prompt quality is nevertheless structured. A displaced point may remain inside an object, whereas a
22 shifted box changes localization and spatial support. Equal numeric noise scales do not imply equal
23 prompt difficulty. Prior work compares boxes, centroids, and random points for SAM [8], adapts with
24 perturbed prompts [9], estimates SAM uncertainty [10], and measures the gap between simulated
25 and human interactions [11]. We therefore measure the realized quality of each channel instead of
26 assuming equal perturbation strength.

27 The paper is therefore organized around robustness evaluation, with Center Color, GrabCut, and
28 the superpixel pipelines used as interpretable probes rather than as claims of a fundamentally new
29 segmentation architecture. We make three contributions: (1) a hash-pinned train/validation protocol
30 that calibrates point and box perturbations to matched observable quality; (2) a full-validation
31 study of three pre-specified questions with paired inference and family-wise error control; and (3)
32 component, class, target-size, and failure-case analyses that translate aggregate scores into practical
33 method-selection guidance.

34 2 Related Work

35 **Interactive segmentation.** GrabCut combines graph cuts and iterative appearance models [2].
36 DEXTR conditions segmentation on extreme points [4]; RITM and SimpleClick improve iterative
37 click-based segmentation across datasets [5, 6]. Human studies show that simulated interactions do
38 not automatically reproduce annotation behavior [11].

39 **Promptable foundation models.** SAM supports point, box, and mask prompts [7]. SAM 2 extends
40 this to streaming video with a memory-based architecture and improved speed [14]. EfficientSAM
41 leverages SAM-guided masked-image pretraining to produce a lightweight ViT encoder that retains
42 most zero-shot performance at a fraction of the cost [15]. HQ-SAM introduces a high-quality output
43 token and global-local feature fusion, improving boundary precision on fine structures without re-
44 training the original encoder [16]. Variational prompting evaluations report that boxes can be stronger
45 than centroid or random points [8]. PP-SAM studies perturbed-prompt adaptation [9]; UncertainSAM
46 formalizes uncertainty estimation [10]; recent work also optimizes point prompts using learned agents
47 [12]. BREPS provides the first systematic benchmark of bounding-box robustness in promptable
48 segmentation, finding up to 30% IoU variation across human annotators [17]. These results establish
49 prompt robustness and model efficiency as active topics. Our narrower contribution is matched-quality
50 full-validation evidence with pre-specified paired inference and a transparent lightweight method
51 with component ablations.

52 **Transparent baselines.** SLIC provides efficient, spatially regular Lab-space primitives [3]. We use
53 SLIC as a representation rather than claim it as a new learned method. Comparison with GrabCut
54 and explicit ablations prevents method complexity from being mistaken for superiority.

55 3 Methods

56 3.1 Task and prompts

57 For each VOC image, the largest connected foreground component across semantic labels 1–20
58 defines one binary target. The clean box is its tight rectangle; the clean point maximizes the Euclidean
59 distance transform inside the target. These prompts use ground-truth geometry and define a controlled
60 benchmark, not a model of unaided human annotation. No method is trained or fine-tuned on VOC
61 validation.

62 3.2 CPU methods

63 Center Color estimates an RGB prototype around the point, thresholds color distance within the box,
64 and applies morphology and point-connected component selection. Robust Superpixel computes
65 SLIC labels and mean Lab/spatial features, estimates point-induced foreground and box-induced
66 background prototypes, combines relative prototype distance with a spatial prior, unions the result
67 with Center Color, and applies the same cleanup. Ablations remove the color seed, spatial prior, or
68 three-box consensus.

69 GrabCut uses only the prompt: outside-box pixels initialize background, inside-box pixels probable
70 foreground, and a point disk sure foreground. Five iterations are used. Eight OpenCV initialization
71 failures are retained and scored as zero.

72 **Adaptive Superpixel.** The original Robust Superpixel uses a fixed SLIC segment count of 280
73 across all images. Adaptive Superpixel replaces this with an image-size-aware count: $n_{\text{seg}} =$
74 $\max(80, \min(500, \lfloor \sqrt{HW} \cdot 2.0 \rfloor))$. Larger images thus receive finer superpixel resolution, while
75 smaller images use fewer, more coherent segments. This preserves a more consistent image-scale
76 granularity across varying resolutions. All other components (color seed, spatial prior, box consensus)
77 are identical to Robust Superpixel, making this a clean one-factor ablation. Target-area stratification
78 is reported only as a secondary diagnostic because target area is not an input to the schedule.

79 3.3 SAM and candidate selection

80 SAM ViT-B is evaluated with point-only, box-only, and point-plus-box prompts. Under noisy point-
81 plus-box input, five extra prompts are sampled locally around the observed prompt without access
82 to the clean prompt. We compare the single noisy prediction, maximum SAM score, mask-IoU
83 consistency medoid, majority vote, and Oracle best-of-six. Oracle uses target IoU and is a descriptive
84 upper bound only.

85 4 Evaluation Protocol

86 4.1 Data separation

87 The tuning split contains 100 VOC train images: five largest-component targets per class, sampled
88 with seed 20260719. The evaluation split contains all 1,449 VOC validation rows. The earlier
89 30-image analysis motivated the hypotheses; because the final evaluation uses the complete validation
90 split, those images necessarily reappear and the two stages are not fully dataset-disjoint. Data
91 manifests contain identifiers and geometry but no image or mask payloads; manifests, sources,
92 algorithm files, SAM source, and the ViT-B checkpoint are SHA-256 pinned.

93 4.2 Prompt calibration

94 On tuning data only, independent deterministic grid searches match point hit rate and box IoU to the
95 same quality targets using 20 trials per target.

Table 1: Tuning-only noise calibration.

Severity	Target	Point scale	Hit rate	Box scale	Box IoU
Mild	0.85	0.1675	0.8480	0.0400	0.8538
Moderate	0.65	0.2725	0.6495	0.1150	0.6509

96 On validation, moderate point hit rate is 0.6391 and box IoU is 0.6514. This difference is reported
97 without post-hoc recalibration.

98 4.3 Hypotheses and inference

99 The primary metric is sample-macro IoU. Five SAM trials are averaged within each sample. H1 states
100 Robust Superpixel exceeds Center Color. H2 states SAM score selection exceeds a single moderate
101 noisy prompt. H3 states matched-quality box noise causes a larger loss than point noise, equivalently
102 point-noise IoU minus box-noise IoU is positive.

103 Each paired effect uses a 20,000-replicate bootstrap interval and a 50,000-permutation sign-flip test.
104 H1–H3 are corrected together by Holm’s method at family-wise $\alpha = 0.05$. Dice, resource use, strata,
105 alternative selectors, and Oracle are secondary or descriptive.

106 5 Results

107 5.1 CPU baselines and ablations

108 H1 is supported: Robust Superpixel improves over Center Color by 0.0640 IoU, 95% CI [0.0585,
109 0.0696], Holm $p = 0.000060$. GrabCut is stronger. Removing the color seed costs 0.1411 IoU and
110 removing the spatial prior costs 0.0217. Removing multi-box consensus changes mean IoU by only
111 -0.00004 , contradicting the exploratory intuition that box voting was important.

112 5.2 Adaptive superpixel resolution

113 This secondary analysis scales the SLIC segment count between 80 and 500 proportionally to $\sqrt{HW}$.
114 The paired mean IoU improvement is $+0.0074$ with 95% bootstrap CI $[+0.0053, +0.0096]$. Per-area-
115 quartile deltas are $+0.0145$ (Q1, smallest), $+0.0059$ (Q2), $+0.0054$ (Q3), and $+0.0037$ (Q4, largest).

Table 2: Complete VOC validation CPU results. Failures are scored as zero.

Method	IoU	Dice	Median ms (all)	Failures
Center Color	0.5404	0.6753	13.1	0
GrabCut point+box	0.6859	0.7751	416.8	8
Robust Superpixel	0.6044	0.7345	199.8	0
Adaptive Superpixel	0.6117	0.7410	199.3	0
no color seed	0.4633	0.5977	187.0	0
no spatial prior	0.5827	0.7152	197.6	0
single box	0.6043	0.7344	188.5	0

Table 3: Secondary comparison of adaptive and fixed SLIC segment counts on VOC validation.

Method	Mean IoU	Mean Dice	Med. latency	Δ IoU
Robust Superpixel (fixed 280)	0.6044	0.7345	199.8 ms	—
Adaptive Superpixel	0.6117	0.7410	199.3 ms	+0.0074

The effect is modest—less than the colour-seed or spatial-prior contributions—and adds no model fit or inference call. Because the schedule observes image dimensions rather than target area, the quartile pattern is descriptive evidence rather than a target-size mechanism.

5.3 Stratified analysis

Table 4 reports representative per-class IoU. Performance is bimodal: structurally simple, large, box-shaped classes (tvmonitor 0.724, sofa 0.720, bus 0.702) exceed 0.70 IoU, while thin-structured classes (bicycle 0.203, motorbike 0.548) fall far below the dataset mean. The nine bicycle samples among the worst 20—23.7% of all bicycles versus 1.3% for cars—confirm that superpixel-based methods fail systematically on wire-like geometries where even the tight bounding box contains substantial background. Figure 2 in Appendix A gives the complete all-method comparison.

Table 4: Per-class IoU for selected classes (full Robust Superpixel).

Best classes	IoU	Worst classes	IoU
tvmonitor	0.7237	bicycle	0.2032
sofa	0.7195	aeroplane	0.5333
bus	0.7023	motorbike	0.5479
cat	0.6704	chair	0.5488
dog	0.6169	bird	0.5916
Sample-weighted mean (20 classes)			0.6044

Effect of target area. All methods improve on larger targets. The largest area quartile exceeds the smallest by 0.064 IoU (Center Color), 0.210 (GrabCut), and 0.118 (Robust Superpixel). The colour-seed ablation shows the steepest area dependence (+0.253 IoU from Q1 to Q4), indicating that colour-based segmentation is most fragile on small regions where reliable prototypes are harder to estimate. GrabCut’s lower mean performance on small targets is consistent with less stable appearance estimation from fewer foreground pixels (Figure 3 in Appendix A); this aggregate degradation is distinct from its eight hard `initGMMs` failures, which are caused by full-image boxes leaving no definite-background pixels.

Failure cases. The 20 worst Robust Superpixel predictions ($\text{IoU} < 0.13$) share three traits: (i) 9 of 20 are bicycles, (ii) median target area is 7,933 px vs. 27,023 px dataset-wide, and (iii) three targets below 500 px produce near-zero IoU. On these difficult samples, Center Color (0.14 IoU) and GrabCut (0.30 IoU) both outperform Robust Superpixel (0.07), suggesting that the superpixel pipeline’s additional structure can be counterproductive when the underlying SLIC segmentation is itself unreliable on thin or tiny objects. Conversely, the best 20 predictions are dominated by large,

uniformly coloured objects (sofas, cats, monitors) with median area 35,687 px. Appendix Figure 4 visualizes representative extremes.

5.4 SAM modality

Table 5: Clean SAM ViT-B prompt modalities.

Prompt	IoU	Dice
Point only	0.5389	0.6235
Box only	0.8479	0.9058
Point+box	0.8491	0.9087

Point+box is only 0.0012 IoU above box-only. The full-validation result refines the exploratory claim: SAM is box-dominated, but box-only is not stronger in aggregate.

Figure 1 compares SAM and CPU methods per class. A striking pattern emerges: SAM point-only underperforms the simple Center Color baseline on several rigid-object classes (tvmonitor: 0.518 vs. 0.668; sofa: 0.411 vs. 0.670; bus: 0.336 vs. 0.577). Under this protocol, the distance-transform point identifies a safe interior location but supplies no object extent. SAM box-only, by contrast, dominates all CPU methods on almost every class (0.924 on tvmonitor, 0.947 on bus), showing that the tight box is much more informative than a single interior point for SAM ViT-B in this benchmark. On bicycle, even SAM box-only reaches only 0.356, consistent with thin structures and substantial background inside the tight box; the experiment does not isolate a dataset or architecture ceiling.

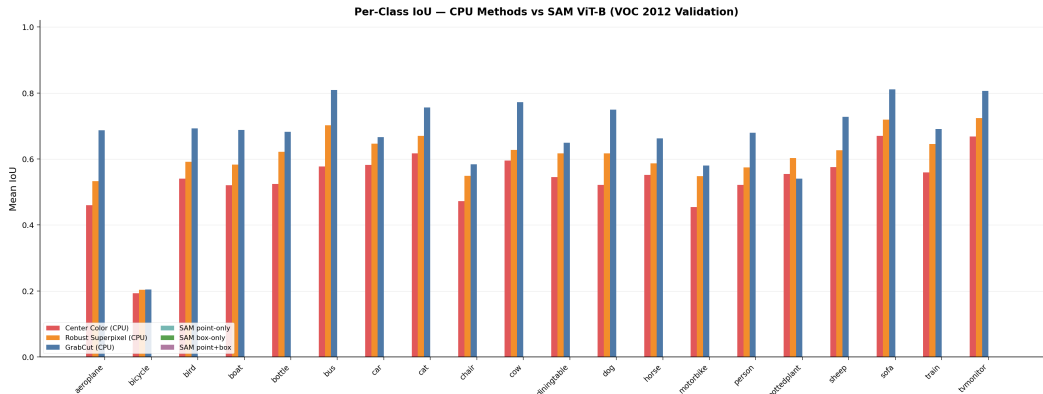


Figure 1: Per-class IoU: CPU methods vs. SAM ViT-B clean-prompt modalities. SAM point-only underperforms CPU methods on rigid, box-shaped classes (tvmonitor, sofa, bus); SAM box-only dominates almost everywhere except bicycle.

5.5 GrabCut failure investigation

The eight GrabCut failures (0.55% of validation samples) share an exact geometric condition: every tight box spans the full image. Our mask initializer labels only pixels outside the box as definite background, so these cases supply no background samples and OpenCV raises the `initGMMs` assertion. Their large median target area (144,517 px vs. 27,023 px dataset-wide) and sofa frequency (3 of 8) are accompanying characteristics, not the established cause. All eight failures are conservatively scored as zero IoU in aggregate metrics. A deployable implementation should detect a full-image box and use a fallback such as Center Color or an inset/background-border initialization.

5.6 Primary hypotheses

Moderate point noise yields 0.8194 IoU and matched-quality box noise 0.7125, supporting H3. Under joint noise, a single prompt reaches 0.6529, score selection 0.6723, consistency medoid 0.6758, vote consensus 0.6710, and Oracle 0.7822. H2 is statistically supported but modest. Consistency medoid was not a primary comparison and remains exploratory; Oracle is not deployable.

Table 6: Pre-specified paired IoU effects with Holm correction.

Comparison	Mean Δ	95% CI	Holm p
H1: Robust Superpixel — Center Color	0.0640	[0.0585, 0.0696]	0.000060
H2: score selection — single noisy	0.0193	[0.0135, 0.0251]	0.000060
H3: point-noise IoU — box-noise IoU	0.1069	[0.0997, 0.1138]	0.000060

Multi-quality sensitivity. In a pre-specified secondary analysis, tuning-only calibration targets observable qualities 0.9, 0.8, 0.7, 0.6, and 0.5; all 1,449 validation samples are then evaluated with 20 trials per target and channel (289,800 SAM inferences). Point-noise minus box-noise IoU is -0.0016 $[-0.0041, 0.0009]$ at 0.9, then 0.0238 $[0.0201, 0.0275]$, 0.0756 $[0.0703, 0.0809]$, 0.1497 $[0.1430, 0.1563]$, and 0.2324 $[0.2246, 0.2402]$. Thus H3’s direction is stable from 0.8 to 0.5 but not distinguishable at near-clean quality. Matching hit rate to box IoU remains an engineering calibration, not perceptual equivalence.

5.7 Practical method selection

Table 7 distills the experimental evidence into actionable recommendations. The choice of method should depend on the target-object profile and available resources, not on aggregate benchmark ranking alone.

Table 7: Method selection guide based on target characteristics and constraints.

Scenario	Recommended method (and why)
GPU available, box prompt feasible	SAM box-only (0.848 IoU; box dominates; point adds <0.002)
GPU available, point-only only	SAM point-only is competitive for cats/cows/birds (0.72–0.81); Center Color is stronger on rigid objects (0.58–0.67 vs. SAM 0.34–0.52)
CPU only, accuracy priority	GrabCut point+box (0.686 IoU); full-image boxes cause 8/1449 initialization failures—fall back to Center Color
CPU only, speed priority	Center Color (13 ms, 0.540 IoU) or Adaptive Superpixel (199 ms, 0.612 IoU)
Bicycles (thin/wire-like)	SAM box-only is the only method exceeding 0.30 IoU; all CPU methods remain below 0.21
Large rigid objects (tv, sofa, bus)	GrabCut or SAM box-only; even Center Color reaches 0.58–0.67
Small targets ($< 12K$ px)	Adaptive Superpixel (+0.0145 over fixed, secondary); avoid GrabCut colour-seed ablation (0.28 IoU)

A deterministic 200-sample ADE20K evaluation, selected after scanning all 2,000 validation rows, provides a bounded transfer observation; Appendix B reports its scope and results.

6 Discussion

Representative evaluation changes the original interpretation. The proposed CPU method is reliably better than its minimal baseline, but not GrabCut; its value lies in transparency and a speed–accuracy operating point. Box localization remains SAM’s principal signal, consistent with prior prompting studies [8] and BREPS [17]; our added evidence is that its greater sensitivity emerges progressively as matched observable quality degrades, while the channels are indistinguishable near 0.9. SAM score selection becomes statistically reliable at full scale, although the improvement is only 0.0193 IoU and requires extra decoder calls.

Stratified analysis reveals systematic weaknesses: bicycles and other thin-structured objects consistently fail ($\text{IoU} < 0.21$) because superpixel segmentation cannot capture wire-like geometry, and small targets (lowest area quartile) suffer across all CPU methods. The colour-seed component is the most area-sensitive (+0.25 IoU Q4–Q1), confirming that colour prototypes degrade on small regions.

191 These per-class and per-size patterns—not just aggregate scores—should guide users in selecting the
192 appropriate method for their target-object characteristics.

193 Negative results are equally informative. Multi-box consensus adds complexity without aggregate
194 benefit. The strongest CPU method is classical GrabCut. On the hardest 20 samples, all CPU
195 methods fail, but Center Color (0.14) and GrabCut (0.30) are more robust than the superpixel-based
196 method (0.07), suggesting that structural priors can become liabilities when the underlying image
197 segmentation is unreliable.

198 **7 Threats to Validity**

199 Ground-truth-derived prompts are more controlled than real user input. Numeric matching of hit
200 rate and box IoU does not make the metrics psychologically equivalent, and neither distribution
201 is fitted to human errors. A privacy-minimizing human-prompt pilot protocol and collection tool
202 are provided, but no participants were recruited and no human result is claimed. The benchmark
203 selects one largest component per image rather than every instance. VOC 2012 and SAM ViT-B are
204 the only full-scale model evaluation. The ADE20K study deterministically samples 200 of 2,000
205 validation rows and evaluates CPU methods only; it improves transfer coverage but is not a complete
206 second-dataset confirmation. Full ADE20K evaluation, larger checkpoints, and learned interactive
207 baselines would improve external validity. The earlier exploratory 30 images reappear inside the
208 complete validation split (2.1%), so H1–H3 are full-scale but not independently preregistered on
209 a disjoint dataset. Adaptive Superpixel and the five-target curve are secondary analyses, and the
210 image-size schedule does not identify a target-size mechanism.

211 **8 Conclusion**

212 Across all 1,449 VOC validation images, Robust Superpixel improves over Center Color but trails
213 GrabCut; a secondary adaptive-resolution comparison yields a small paired gain; and SAM’s box
214 channel dominates, with point-only underperforming CPU baselines on several rigid classes. Matched-
215 quality box noise causes substantially larger loss than point noise in H3; a five-target curve localizes
216 this asymmetry to quality 0.8 and below, with no distinguishable difference near 0.9. SAM score
217 selection offers a modest but Holm-significant gain. Bicycles and small targets are systematic failure
218 modes, while GrabCut’s eight hard failures arise from full-image boxes. Together, these results
219 provide a reproducible, stratified account of prompt-channel robustness and practical trade-offs.

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247 A Extended stratified analysis

248 The main paper reports the numerical conclusions required for the hypothesis tests. This appendix
 249 provides the corresponding full-resolution diagnostic plots.

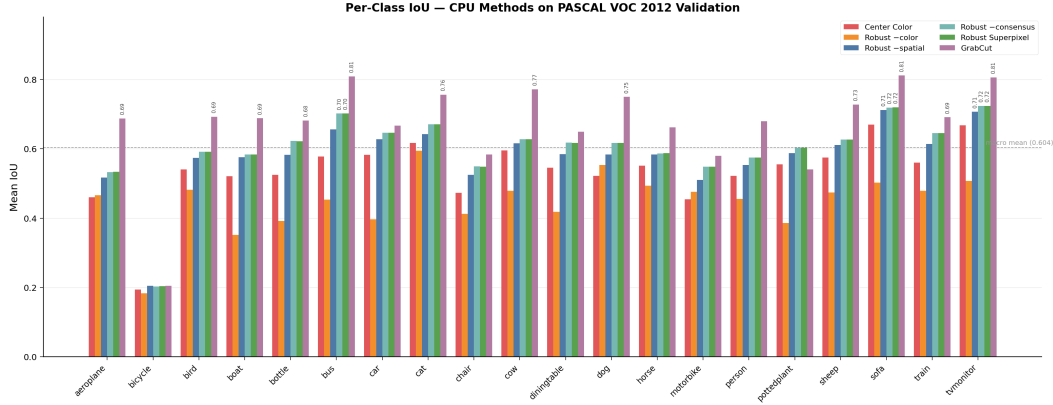


Figure 2: Per-class mean IoU for all CPU methods on VOC 2012 validation. Bicycle is systematically the most difficult class (all methods < 0.21 IoU). GrabCut dominates except on pottedplant, where it underperforms due to GMM initialization sensitivity on multi-colour small-pot scenes.

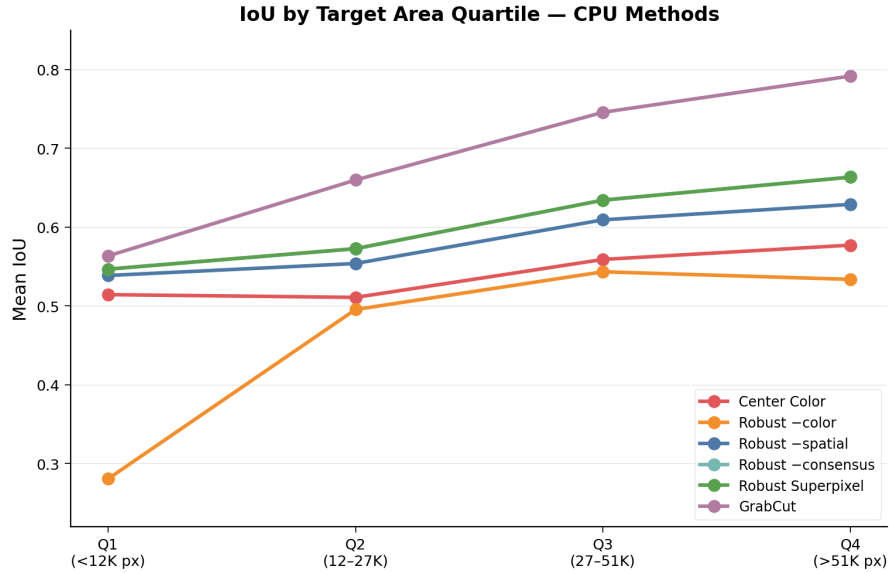


Figure 3: Mean IoU by target area quartile. All methods improve with area; the colour-seed ablation (Robust-color) shows the steepest climb, reflecting unreliable colour prototypes on small regions.

250 B Bounded ADE20K transfer observation

251 We scan all 2,000 rows of an ADE20K validation mirror [13], then use seed 20260720 to select
 252 200 rows without replacement for CPU evaluation. All rows pass the 256-px minimum-area filter.
 253 Samples come from the laurent/ADE20K Hugging Face mirror; targets and clean prompts follow
 254 the VOC protocol. The manifest, selected identifiers, per-sample metrics, and hashes are released
 255 without images or masks.

256 The IoU ranking is GrabCut $>$ Adaptive Superpixel $>$ Robust Superpixel $>$ Center Color. Removing
 257 the color seed costs 0.1989 IoU and removing the spatial prior costs 0.0332, while the single-

Robust Superpixel — Failure & Success Cases (Error Maps)

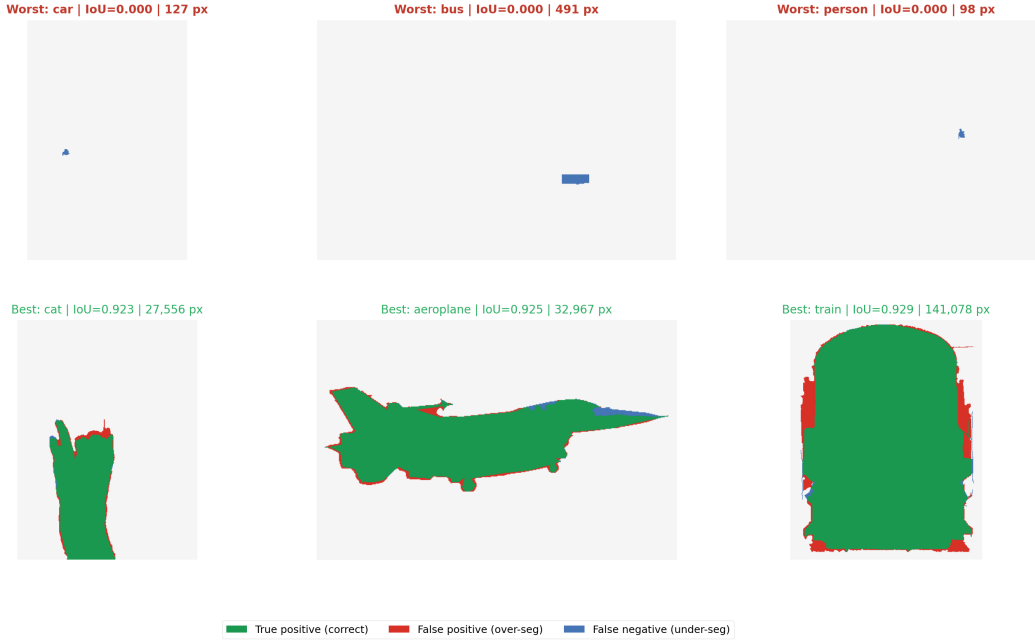


Figure 4: Error maps for three worst and three best Robust Superpixel predictions. Green = true positive, red = false positive, blue = false negative. Worst cases (top row) are dominated by thin structures and small targets; best cases (bottom row) are large, uniformly coloured objects.

Table 8: CPU methods on a deterministic 200-sample ADE20K subset.

Method	Mean IoU	Mean Dice	Med. latency
Center Color	0.6350	0.7541	23.9 ms
GrabCut point+box	0.7383	0.8169	1901.7 ms
Robust Superpixel	0.7150	0.8178	362.9 ms
Adaptive Superpixel	0.7288	0.8278	361.7 ms
no color seed	0.5162	0.6567	339.2 ms
no spatial prior	0.6819	0.7903	360.6 ms
single box	0.7159	0.8184	338.3 ms

box variant differs from the multi-box default by only +0.0008. Adaptive resolution improves descriptively by 0.0138 IoU over the fixed schedule. These patterns agree with the VOC component analysis, but this bounded secondary sample does not replace a complete cross-dataset evaluation.

Coverage. The complete 2,000-row stream was scanned and all rows were eligible; 200 seeded rows (10%) were evaluated with seven CPU methods, producing 1,400 metric rows with no method failures. The same resume-safe runner supports full evaluation, but the remaining 1,800 rows were not run.